DANIEL LANDI

AI & Infrastructure Engineer | Local LLMs · Cloud · Systems Programming
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PROFESSIONAL SUMMARY

AI and infrastructure engineer with 25+ years of experience operating mission-critical telecom systems across seven countries, now building AI applications and local-LLM tooling. Recent work includes a C inference runtime on llama.cpp designed for 8 GB, CPU-only hardware; a self-correcting LLM activity generator containerized with Docker and deployed on Railway; typed-decision LLM demos; and serverless AWS pipelines with Bedrock and Textract. Completing the 42 São Paulo systems-programming curriculum. Brings carrier-grade operational discipline — SLAs, failure-first design, documentation someone else can act on at 3 AM — to AI systems that have to keep working.

CORE SKILLS

AI & LLM engineering: Local inference with llama.cpp (C API, GGUF, sampling and decoding); LLM application design (prompt pipelines, self-critique loops, structured and typed outputs); Groq-hosted Llama 3; AWS Bedrock and Textract; IBM Bob 2.0 CLI

Programming & tooling: C (42 Norm), Python, Node.js/Express, Bash, Git/GitHub, Makefiles, Docker, manual memory management

Cloud & infrastructure: AWS: Lambda, API Gateway, DynamoDB, S3, CloudFront, WAF, SQS, SNS, CloudWatch, IAM, Route 53; Terraform (environment audit); Linux/Unix (Pop!_OS, Solaris, HP-UX); high-availability network architecture

Operations & process: ITIL, eTOM, SLA and uptime management, NOC operations, SOPs and technical documentation

AI & SOFTWARE PROJECTS

Self-Correcting AI Teaching Activity Generator | IBM Bob 2.0 Hackathon *Sep 2026*

- Built and deployed, during a 48-hour hackathon, a Node.js/Express web app that generates hands-on, teacher-editable learning activities adapted to a classroom's resource level, with a second-pass self-critique loop that reviews and regenerates output against quality checks.
- Integrated the Bob CLI through child processes (no public HTTP endpoint), resolved stream-handling hangs, added print-ready PDF and flowchart views, and containerized the app with Docker for a permanent Railway deployment. Validated against classroom activities I had already designed at CIEBP. Code: github.com/drlandi/ibm-bob-activity-generator

uLLAMA — Local LLM Inference Runtime in C | Personal project *Sep 2026 – Present*

- Wrote a C runtime on the llama.cpp API that loads GGUF models and runs an interactive session with streamed output, configurable sampling (top-k, top-p, temperature) and live tokens-per-second measurement; designed to fit 8 GB RAM on CPU only, written under the 42 Norm.
- Repository scaffolded for domain-specific fine-tuning datasets and an offline code-generation harness (planned). Code: github.com/drlandi/ullama

Typed Decisions vs. Prose — LLM Triage Demo | Personal project *Sep 2026*

- Built and deployed a side-by-side support-ticket triage demo comparing typed-decision output with free-text generation, following a published model I/O contract (simulated locally while the API is in closed beta); shipped English and localized Brazilian Portuguese versions.

ClaimsCopilot | Personal project *2026*

- Developed an AI insurance-claims assistant using Groq-hosted Llama 3 for natural-language claim analysis.

Intelligent Insurance Claims Triage ("Elementum") | AWS Hackathon *2026*

- Built a serverless claims-intake pipeline on S3, Lambda and DynamoDB, with Textract/Bedrock document-intelligence integration (mocked due to account verification constraints).

Digital Resilience Cloud Architecture | EDN Hackathon "Future Builders" (Escola da Nuvem × AWS) *Sep 2026*

- Designed, built and live-tested a failover-resilient serverless architecture: WAF and CloudFront with dual-origin failover (S3 in sa-east-1 and us-east-1), API Gateway with direct-to-SQS integration, Lambda worker, DynamoDB, CloudWatch/SNS. Validated end-to-end with real submissions and audited a teammate's Terraform-provisioned environment against requirements.

"Gems & Landi" AWS Cloud Labs Documentation Series | Personal project *2026 – Present*

- Author an in-depth documentation series (Markdown/Obsidian) for hands-on AWS labs in networking, security and serverless, emphasizing design rationale over copy-paste steps.

Systems Programming Coursework | 42 São Paulo — Software Engineering Cadet *2025 – Present*

- libft (43 C standard-library functions), get_next_line (static memory; fixed malloc-failure edge cases and two leaks), push_swap (co-developed adaptive sorting for large data sets), ft_printf (variadic arguments).

PROFESSIONAL EXPERIENCE

Specialist Teacher (Professor Especialista) | CIEBP — São Paulo State Public School *Feb 2023 – Present*

- Design and teach activity series that bring computational thinking (unplugged programming) and AI literacy to classrooms with limited access to technology; building AI-assisted tooling to generate and quality-check classroom activities.

Electrical Engineering Consultant | TAUPO – Consultoria Ambiental *Nov 2015 – Dec 2021*

- Delivered low-voltage building electrical projects and solar-energy projects in rural areas of northwestern Minas Gerais.

Maintenance Operations Coordinator | SEICOM (Vivo network, Espírito Santo) *Nov 2014 – Mar 2015*

- Coordinated preventive and corrective maintenance of RF and infrastructure for Vivo's cellular sites; managed the technical and administrative team, reporting and contract targets.

Technical Education & Curriculum Leadership | UFES, Multivix, Master Grupo Educacional *2010 – 2013*

- Lecturer, Universidade Federal do Espírito Santo, São Mateus (Dec 2011 – Sep 2013): thermodynamics, strength of materials and solid mechanics for Petroleum, Computer, Chemical and Production Engineering. Lecturer, Multivix (2011 – 2013): physics and solid mechanics for Mechanical and Civil Engineering.
- Technical Course Coordinator, Master Grupo Educacional (Nov 2010 – Jun 2013): led Electrotechnics, Industrial Automation, Mechanics and Oil & Gas programs — operating procedures, teaching labs, course materials, hiring and managing faculty.

Senior Global Systems & Operations Consultant | Ericsson Telecomunicações Ltda. *Nov 2005 – Jul 2010*

- International specialist on GSM and 3G (WCDMA) projects for carriers in six countries outside Brazil, designing and implementing operational processes under eTOM and ITIL.
- Zain (Nigeria): managed the Field Operations sub-project and new operating procedures under eTOM. Hutchison 3G (UK, Australia) and Etisalat (Egypt): contractual assessment and development of operational processes.
- Nawras (Oman): network configuration management project. EGS (Brazil): field-operations management system. Telkomsel and Natrindo (Indonesia): operational-process implementation.

Network Operations Center Engineer | Nextel S.A. *Apr 1998 – Oct 2005*

- Preventive and corrective maintenance of Nortel GSM switches, Tekelec signaling transfer points, messaging, voicemail, prepaid and Motorola base-station management systems; administered Unix systems (Solaris, HP-UX).
- Network performance analysis, systemic analysis reports, GSM switch software upgrades; authored operating procedures.

Network Operations Center Engineer | Telesp Celular S.A. *Apr 1993 – Apr 1998*

- Maintained NEC, Ericsson and Nortel cellular switches; performance reporting, drive tests and wireless network optimization; designed and implemented a LAN on Windows NT 4.0.

Telecommunications Technician | Bastec S/A and Telepatch S/A *1991 – 1993*

- PABX and microwave-link maintenance for a banking network; acceptance testing and commissioning of VHF/UHF transceivers.

EDUCATION

- **42 São Paulo — Software Engineering Cadet** (2025 – present): intensive, project-based, peer-to-peer curriculum in C, algorithms and Unix systems programming
- **Pedagogical Formation Course in Mathematics** — Universidade Pitágoras Unopar (2019 – 2020)
- **B.S., Electrical Engineering** — Faculdade de Engenharia de São Paulo (FESP), 2002
- **Electronics Technician** — Escola Técnica de Eletrônica Francisco Moreira da Costa, Santa Rita do Sapucaí, MG, 1990

TRAINING & CERTIFICATIONS

- ITIL Service Management — CompanyWeb (2007, 2008); eTOM Distilled — TM Forum
- Ericsson (2006): WCDMA System Overview + CPP Node Features, GPRS System Survey, ATM & IP over ATM, Datacom and IP Fundamentals
- Digital Ultrix system administration and utilities (1994); Windows NT 4.0 administration (1997 – 1998)

LANGUAGES

Portuguese (native) · English (fluent) · Spanish (basic)